

2.4 Data Flow Lifecycle & Core Use Cases

Primary Use Case: Student Leave Request Processing

- **Primary Actor:** Student (`x_scms.student`)
- **Pre-conditions:** Student is authenticated on the portal and has available leave balance.
- **Execution Flow:**
 1. Student accesses the Service Portal and opens **Submit Leave Request**.
 2. Student selects Leave Type, Start Date, and End Date.
 3. Client Script executes validation logic to verify date constraints.
 4. Form submission creates a record in `x_scms_leave_request` with State = `Requested`.
 5. Flow Designer initiates an approval task assigned to the Faculty / HOD.
 6. Upon HOD approval, a server-side Business Rule recalculates and updates the student's leave balance quota.
 7. The Notification Engine dispatches an automated confirmation email to the student.

Main Success Scenario: Data Flow Execution

Step #	User Action / System Trigger	Data Component	Technical Execution & Logic
1.0	Student accesses Submit Leave Request on Portal.	Service Catalog (Record Producer)	Form renders; <code>sys_id</code> of the logged-in student is captured into a reference field.
2.0	Student selects <code>Leave Start Date</code> and <code>Leave End Date</code> .	Client Script (<code>onChange</code>)	Script validates date inputs: <code>End Date >= Start Date</code> . An error is displayed dynamically if invalid.

3.0	Student clicks 'Submit'.	Business Rule (before insert)	Server-side validation confirms final date constraints and performs a preliminary leave balance check to prevent over-drawing.
4.0	Record is inserted into the database.	Transactional Table (x_scms_leave_request)	New record is created with State set to Requested .
5.0	Record creation triggers automation.	Flow Designer (Triggers Workflow)	Flow initiates. It retrieves the student's Faculty Advisor and HOD dynamically.
6.0	Approval is requested.	Flow Designer (Stage 1 Approval)	Workflow creates an approval record and assigns it to the Faculty Advisor. State updates to Faculty Review .
7.0	HOD Final Approval.	Flow Designer (Stage 2 Approval)	Upon Advisor approval, the HOD receives an approval request. State updates to HOD Review .
8.0	HOD Approves the Request.	Flow Designer (Success State)	The main request state updates to Approved .

9.0	Execute Post-Approval Logic.	Flow Designer (Actions Group)	Workflow executes: 1. Updates <code>x_scms_leave_request</code> record status. 2. Triggers Business Rule to Recalculate & Update the student's master profile balance in <code>x_scms_student</code> .
10.0	Final Notification.	ServiceNow Notification Engine	Flow triggers an event that sends an automated email update to the Student confirming the approval and final state.